A Split-Readout Dual-Stream Vision Agent

Routing salience to the actor and a top-down gate to the critic in cued, value-weighted change detection

2026

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Abstract

How an agent routes perceptual evidence to its policy versus its value estimate shapes what it can learn. We study a compact recurrent vision agent, trained only by reinforcement on a Posner-style cued change-detection task, whose perceptual core is a pair of parallel single-head attention streams with a deliberately split read-out: a bottom-up salience stream — which keeps the raw image on its residual path and attends from the image into both a fast sensory memory and a slow deep memory — drives the actor, while a top-down stream — whose residual is the deep memory, which the current image can only re-gate — drives the critic. This is a minimal change from a sibling model in which both streams are concatenated and read jointly by both heads, and it produces a strikingly different agent. Where the joint-readout sibling ignores the spatial cue entirely, the split-readout agent recovers the central behavioural signature of spatial attention: validly cued changes are detected more accurately and faster than invalidly cued ones, and — the signature the sibling lacked — this benefit scales with the cue's displayed reliability, growing from a few points at the least reliable cue to twenty points at a fully valid one. Using a faithful, bias-injectable reconstruction of the two streams we show why: routing the grounded salience read to the actor lets the cue's spatial prior, which that stream carries, reach the policy directly. We then map each stream's attention, decode the task variables held in each memory, and causally perturb the streams — confirming a clean architectural dissociation that the split read-out makes almost inevitable: salience attention moves the agent's decision, top-down attention moves its value estimate and uncertainty. We close on a well-calibrated, value-graded critic. The result is a concrete demonstration that where a representation is read out, not just what it encodes, determines which behavioural signatures an agent expresses.

1. Introduction

Detecting change in a cluttered scene requires holding the recent past, comparing it to the present, and deciding under time pressure whether the difference warrants action. Attention shapes this process: a spatial cue that predicts where a change is likely improves accuracy and speeds responses at the cued location, and the size of that benefit grows with the cue's reliability. When outcomes carry different rewards, attention is further pulled toward the valuable ones.

A natural question for a learned agent is not only whether it forms representations of the cue, the change, and the value, but where in its architecture those representations are read out — and whether that routing decides which behavioural signatures emerge. We address this with a deliberately controlled comparison. The agent studied here runs two parallel cross-attention streams on each frame, both querying with the current image:

- a salience stream that keeps the raw image on its residual and attends into memory — here, into both the fast sensory memory and the slow deep memory — producing a grounded read that combines the present image with what is stored at each location;
- a top-down stream that keeps the deep memory on its residual, so the current image can only re-gate the stored readout — a strict, expectation-driven bottleneck.

The single design choice we examine is the read-out. In a sibling model the two stream outputs are concatenated and fed jointly to both the policy (actor) and value (critic) heads. Here we instead split them: the salience output drives the actor, and the top-down output drives the critic. Each head also uses a single attention head per stream rather than eight. Everything else — the task, the environment, the recurrent memory, the reinforcement-learning trainer — is identical. The question is what this routing does to behaviour and to the internal computation.

The answer, developed below, is that the split routing changes the agent's behavioural phenotype qualitatively. We proceed in five steps: (1) we establish the behavioural signatures, including a reliability-scaled spatial cueing benefit absent from the joint-readout sibling; (2) we map the spatial and temporal allocation of attention in each stream; (3) we decode the task variables linearly held in each memory and stream output; (4) we causally perturb each stream's attention to identify which carries the decision and which the value; and (5) we characterise the value function. Every analysis rests on a reconstruction of the model's attention verified to reproduce the trained network to numerical precision, so the causal manipulations act on the real circuit.

1.1 Background and related work

The task is a reinforcement-learning rendering of the cued change-detection paradigm; the behavioural phenomena of interest — improved sensitivity and faster responses at validly cued locations, scaling with cue reliability, and modulation by reward — are robust findings in human and animal attention research. Architecturally the agent belongs to a family of recurrent attention networks in which a transformer-style attention is wrapped in a recurrence so the keys and values are the network's own evolving memory. The specific manipulation studied is the routing of two parallel streams to the actor versus the critic, which — as we show — is sufficient to switch on a behavioural signature that the same streams, read jointly, do not express.

2. Methods

2.1 The change-detection environment

Each trial is a 29-step movie of 50×50 RGB frames. At $t = 1$ a coloured cue appears in one of two locations; the ring's completeness encodes the cue's reliability (proportion $\in \{0.25, 0.5, 0.75, 1.0\}$) and the cue's colour encodes the reward on a correct hit (red $= 5$, green $= 3$, blue $= 1$). From $t = 3$ four oriented Gabor patches are shown, one per quadrant, jittered by orientation noise each frame. On half of trials, at a change time drawn uniformly from $t \in [11, 25]$, one Gabor's orientation steps by $\Delta\theta \in [-64°, 64°]$ for the rest of the trial; otherwise no change occurs. The cue is valid with probability equal to its displayed proportion. The agent emits wait or press each step; pressing before the change ends the trial with zero reward, pressing at/after the change on a change trial yields the cue-colour reward, and correctly withholding through a no-change trial also yields it. Reaction time is the number of frames between change and press, defined on hits.

2.2 Architecture

Each frame is cut into a 10×10 grid of 5×5 patches and embedded into $N = 100$ tokens of width $d = 128$. Writing X for the patch tokens and H_1, H_2 for two per-token recurrent memories carried from the previous frame, the dual-stream encoder computes, with a single attention head each,

$$Z_{\text{sal}} = X + \text{attn}\big(Q{=}X,\ K{=}V{=}[H_1 \,\|\, H_2]\big) + \text{FFN}(\cdot), \qquad \text{(salience; residual } X)$$

$$Z_{\text{td}} = H_2 + \text{attn}\big(Q{=}X,\ K{=}V{=}H_2\big) + \text{FFN}(\cdot), \qquad \text{(top-down; residual } H_2)$$

where the salience stream reads a source-tagged concatenation of both memories (2N keys) and the top-down stream reads the deep memory only (N keys). The memories update by two per-token LSTM cells: $H_1 \leftarrow \text{LSTM}_1(X)$ (written from the raw image) and $H_2 \leftarrow \text{LSTM}_2(Z_{\text{sal}})$ (from the salience output). The split read-out is the defining feature: the actor reads only Z_{sal} and the critic reads only Z_{td}, each through a stack of strided 1-D convolutions over the token axis followed by a linear map (the actor to two wait/press logits, the distributional critic to 51 quantiles per action). The full model has 1,015,656 parameters. Training is a policy-gradient actor loss with a distributional quantile critic and prioritised episode replay; a short burn-in forces waiting while the critic warms up. No part of the training signal specifies where to attend or when to press.

2.3 Faithful attention extraction and intervention

Every mechanistic analysis is built on a hand-written reconstruction of the dual-stream step that reproduces each stream's single-head cross-attention from the network's own projection weights, returns the full attention tensors (salience $(B, 1, N, 2N)$, top-down $(B, 1, N, N)$), and allows an additive bias to be injected into either stream's pre-softmax logits before the softmax. In evaluation mode this reconstruction reproduces the stock network's two stream outputs, both memories, and — through the split read-out — the actor logits and critic quantiles to a maximum absolute deviation below 10^{-6} at zero bias. Because the image queries memory, the spatially-meaningful attention map is the weight each memory position receives, averaged over the image queries; for the salience stream, whose keys are $[H_1 \| H_2]$, we sum the two memories' contributions per position so both streams yield a 10×10 map aligned to the four Gabor quadrants.

2.4 Behavioural, decoding, causal, and value analyses

We measure behaviour by greedy roll-outs on forced-condition trials, computing hit rate, premature-press rate, and reaction time as functions of change magnitude, cue validity (within each displayed reliability), and cue value. We characterise representations by cross-validated linear decoding of each task variable from each memory and stream output at every timestep, against shuffled-label controls, comparing token-mean against per-quadrant pooling. We establish causal roles by sweeping an additive bias over each stream's keys (globally and per quadrant — for the salience stream, separately over its H_1 and H_2 key blocks) and reading the change in the decision versus in value and uncertainty. We assess the critic by comparing predicted values to realised Monte-Carlo returns and tracking its distributional spread against evidence. All analyses use a single frozen snapshot of the trained agent.

3. Results

All results are from a single frozen snapshot of the trained agent. Under its greedy policy on randomly-sampled trials it detects changes well (hit rate 0.80 on change trials), almost never presses prematurely (premature rate 0.01), responds quickly (median two frames), and earns near-ceiling reward (2.63). On this competent agent we examine the behavioural and internal signatures of the split read-out.

3.1 Behavioural signatures: a reliability-scaled spatial cueing benefit

Detection is a graded function of the physical signal — as $|\Delta\theta|$ grows from 2° to 64° the hit rate climbs from near zero to ceiling with a half-maximal threshold near 14°, and reaction time falls from seven frames at threshold to one or two for large changes (Figure 1, left). The decisive result, however, is the spatial cueing benefit. Comparing changes at the cued location (valid) against an uncued location (invalid) with the cue fixed and the location controlled, valid trials are detected more accurately than invalid ones throughout the threshold region, and — the central finding — this benefit scales with the cue's displayed reliability (Figures 2–3). At the near-threshold magnitude $|\Delta\theta| = 14^\circ$, the valid-minus-invalid hit difference is $+0.092$, $+0.132$, $+0.136$ and $+0.208$ for displayed reliabilities of 0.25, 0.5, 0.75 and 1.0 respectively: a monotonic growth from a nine-point benefit at the least reliable cue to a twenty-one-point benefit at a fully valid one. The fully-valid case is the cleanest (Figure 4): a perfectly reliable cue confers its largest advantage, exactly the ordering a reliability-reading observer should show. Reaction time tells the same story, valid responses being faster than invalid ones through the threshold region. Cued accuracy also rises directly with reliability when measured on cued trials alone ($0.68 \to 0.70 \to 0.75 \to 0.75$ across the four proportions).

This reliability-scaled cueing benefit is the behavioural signature that a joint-readout variant of this architecture — in which both stream outputs are concatenated and read by both heads — entirely fails to express, detecting validly and invalidly cued changes equally well at every reliability. The only difference between the two is the read-out routing; routing the grounded salience stream to the actor is what lets the cue shape the policy (Sections 3.2–3.4 trace how).

Value modulates behaviour as well, but gently and monotonically. Holding the change near threshold and varying only the cue colour, the agent detects 80% of red ($v = 5$), 75% of green ($v = 3$) and 71% of blue ($v = 1$) changes (Figure 1, right) — a graded value ordering in which low-value trials are detected slightly less often but not abandoned. The agent's dominant behavioural modulation here is spatial-and-reliability-driven, with value a secondary graded factor.

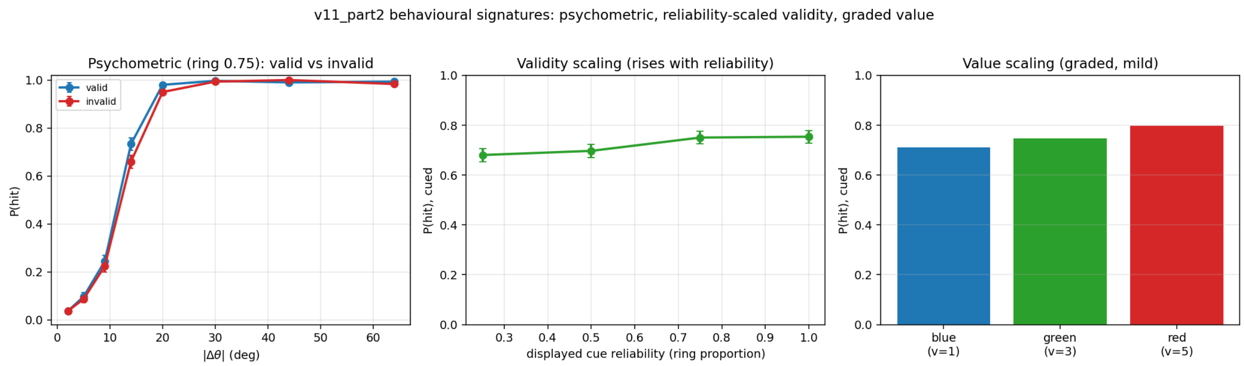

Figure 1: Behavioural signatures. Left: hit rate (solid) and median reaction time (dashed) versus change magnitude for valid vs invalid cues at ring 0.75 — a graded psychometric/chronometric function with valid above invalid through the threshold region. Middle: cued hit rate rises with displayed cue reliability. Right: a graded, mild value effect (red $>$ green $>$ blue), with low-value trials still detected.

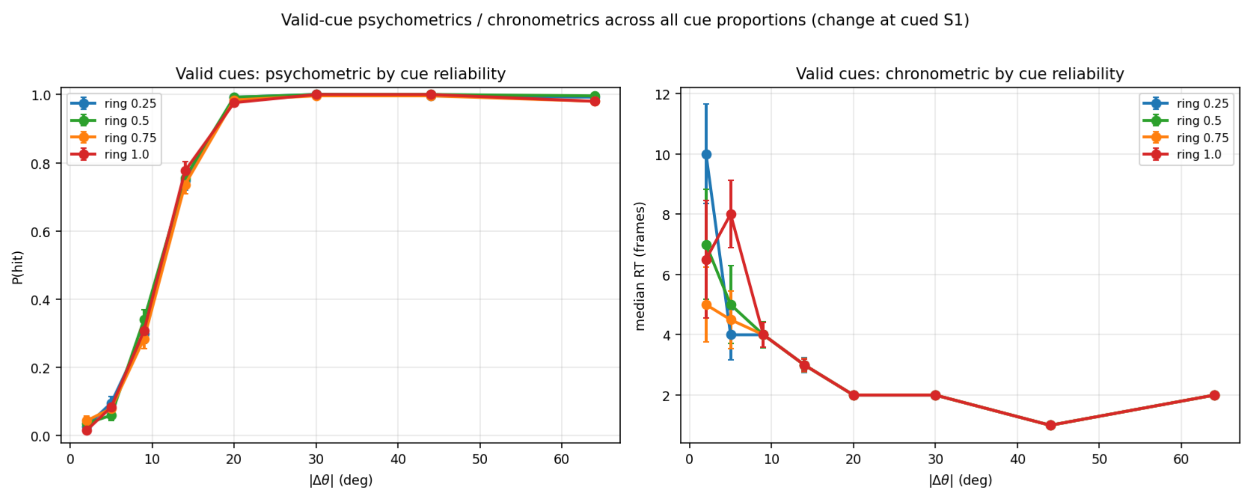

Figure 2: Valid-cue psychometric (left) and chronometric (right) for every displayed reliability (ring 0.25–1.0; change at the cued quadrant S1). Sensitivity is high at all reliabilities; the curves separate from their invalid counterparts (Figure 3) by an amount that grows with reliability.

3.2 How attention is allocated across the two streams

Because the image queries memory, the spatially-resolved quantity is how much attention each memory position receives, averaged over the image queries and laid on the 10×10 grid (for the salience stream, summing its H_1 and H_2 key blocks per position). Read out per quadrant over the trial across all eight cue conditions, the two streams allocate attention differently and in a way that matches their read-out targets (Figures 5–7), with the change held at S1.

The top-down stream — which feeds the critic — tracks the cued location. Through the pre-change period its attention is elevated on the cued quadrant: on S1 when the cue points left (valid; ≈ 0.36) and on S4 when it points right (invalid/cue-right; ≈ 0.39), following the cue rather than the change

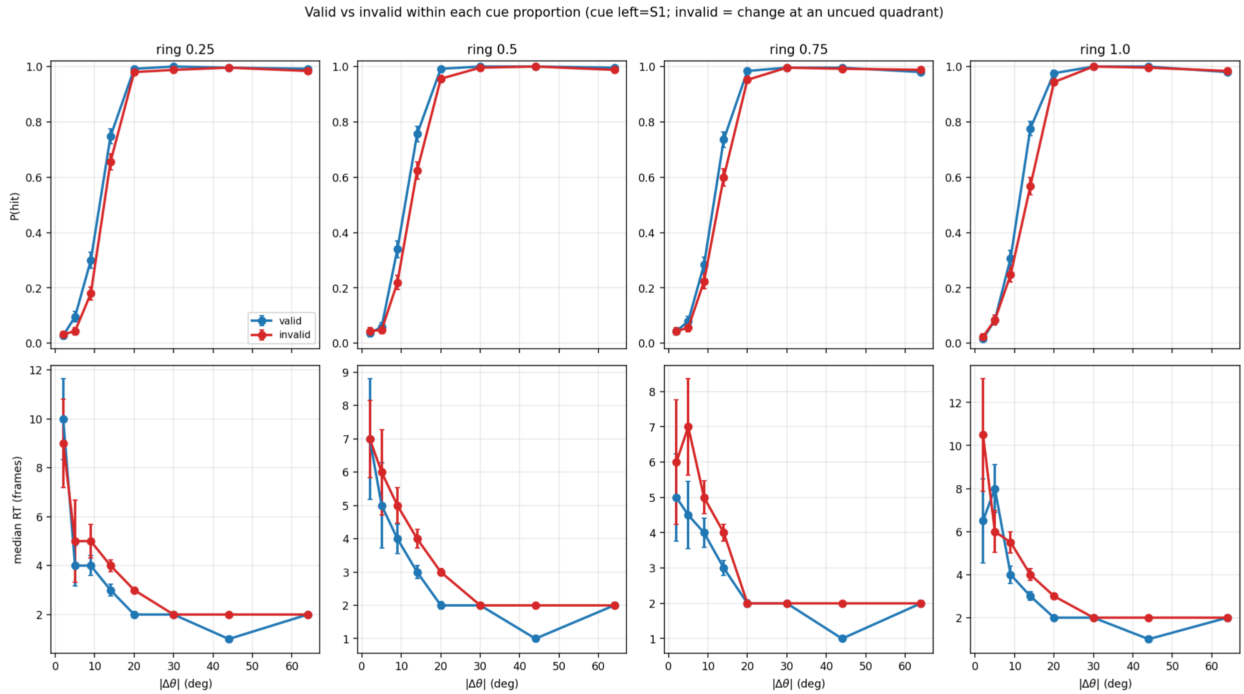

Figure 3: Valid (blue) versus invalid (red) within each displayed reliability (columns: ring 0.25–1.0; top P(hit), bottom median RT). The valid advantage is present at every reliability and widens from left (ring 0.25, $+0.09$ at threshold) to right (ring 1.0, $+0.21$) — a reliability-scaled spatial cueing benefit.

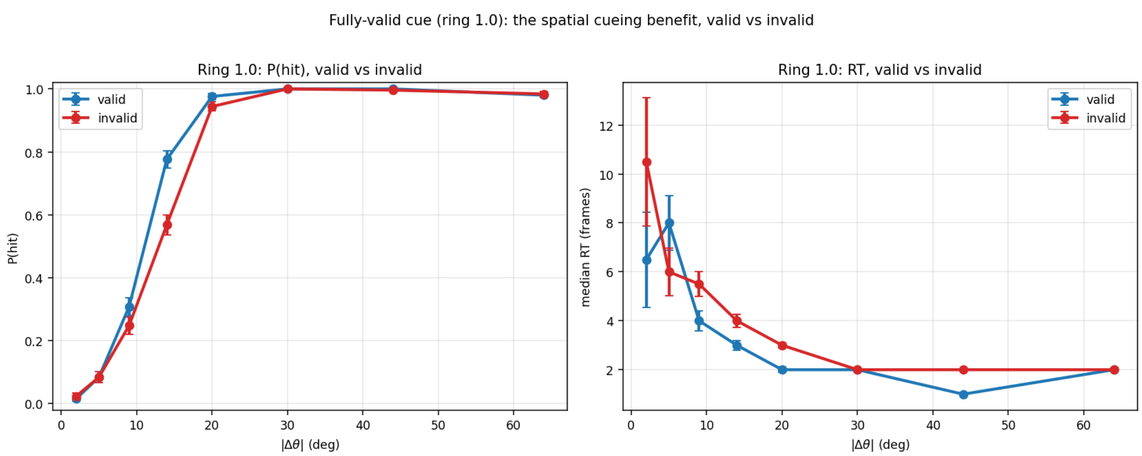

Figure 4: The fully-valid cue (ring 1.0): hit rate (left) and reaction time (right), valid versus invalid. A perfectly reliable cue confers the agent's largest spatial benefit.

(Figures 5, right; 7). It responds only weakly to the change itself. So the cue's spatial prior is carried, in attention, by the stream that drives value.

The salience stream — which feeds the actor — is change-responsive. Its attention is more distributed before the change and then rises on the changed quadrant S1 after onset (to ≈ 0.44), in both valid and invalid conditions, and it is far more sharply peaked than the top-down stream (single-position weights near 1.0 versus a diffuse top-down maximum near 0.27). The salience read-out the actor sees is therefore dominated by the current change, which is consistent with the agent's quick, change-driven pressing.

This raises the question the next two sections answer: if the cue's spatial prior is most visible in the top-down attention (which feeds the critic), how does the cue produce a benefit in behaviour (the actor)? The resolution is that the prior also lives in the memory content that the salience stream reads, not only in attention weights — which the decoding (Section 3.3) and causal (Section 3.4) analyses make precise.

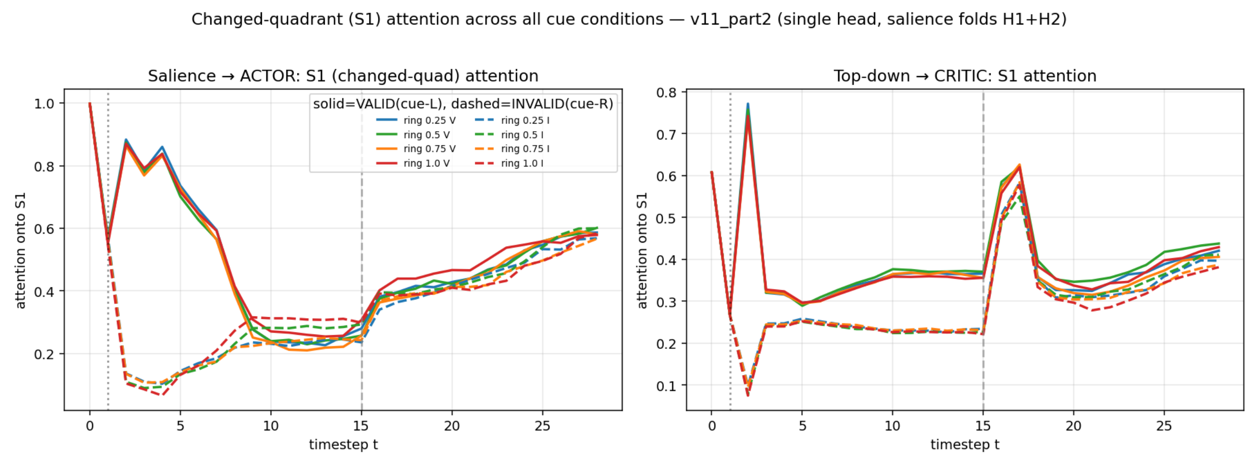

Figure 5: Changed-quadrant (S1) attention over the trial for all eight cue conditions (colour = displayed reliability; solid = valid/cue-left, dashed = invalid/cue-right). Left (salience → actor): S1 attention is change-driven, rising after the change in all conditions. Right (top-down → critic): S1 attention is higher for valid (cue-left, solid) than invalid (cue-right, dashed) through the pre-change period — the cue prior.

3.3 What the recurrent memories and stream outputs encode

Cross-validated linear decoders (against shuffled controls) show that the agent registers the cue and the clock perfectly and holds the change spatially (Figure 8). The cue's colour (hence value) and its reliability are both decodable at ceiling (1.00 balanced accuracy, and reliability also $R^2 = 1.00$ as a continuous regressor) from cue onset, and absolute trial time is decodable at $R^2 = 0.99$. The changed location decodes at 0.83 (four-way, chance 0.25) — but, as in spatial readouts generally, only from per-quadrant-pooled representations; the token mean of every representation decodes location near chance (≈ 0.45). It is recovered best from the per-quadrant sensory memory H_1 (0.83) and the per-quadrant top-down output Z_{td} (0.75), and only weakly from the deep memory's token mean — the spatial "what changed where" lives in the image-written memory and in the critic's stream. Change magnitude and onset are moderately decodable ($R^2 = 0.50$ and 0.62).

The decoders also frame the central puzzle of this agent. The cue's reliability is encoded perfectly — exactly as it is in the joint-readout variant that shows no cueing benefit. Decodability is therefore

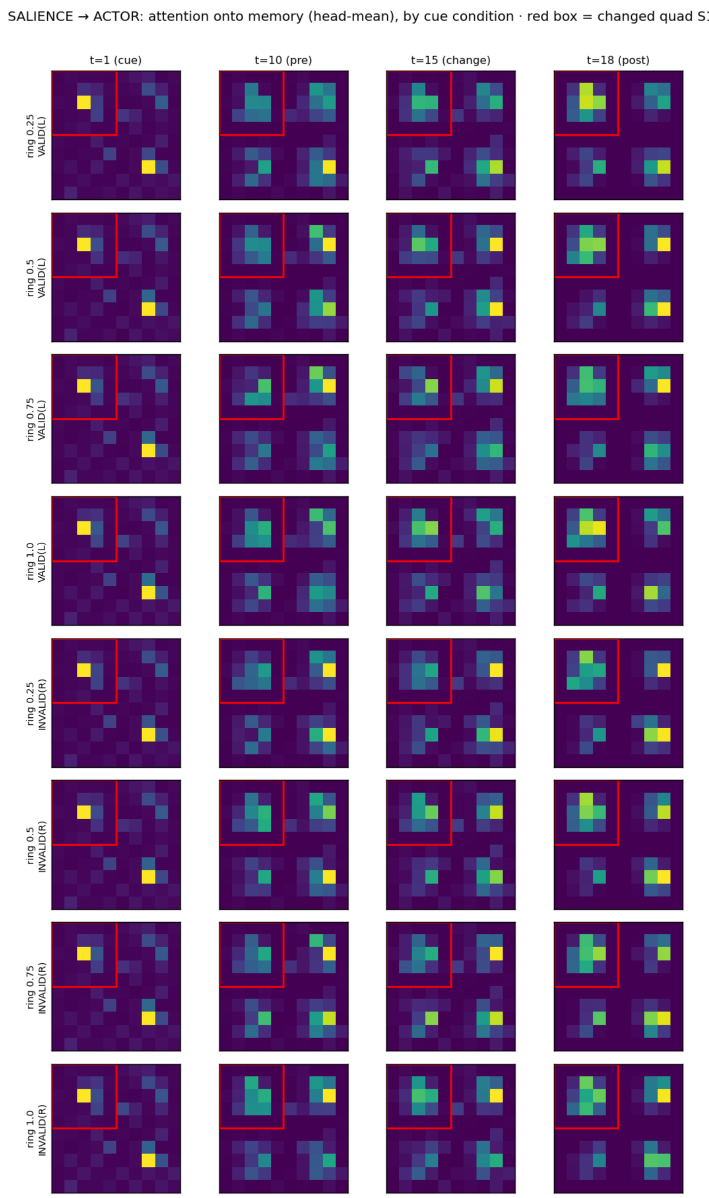

Figure 6: Salience-stream spatial attention maps (head-mean) for every cue condition at four timepoints (red box = changed quadrant S1). The salience read is sharply peaked and becomes change-aligned after $t = 15$.

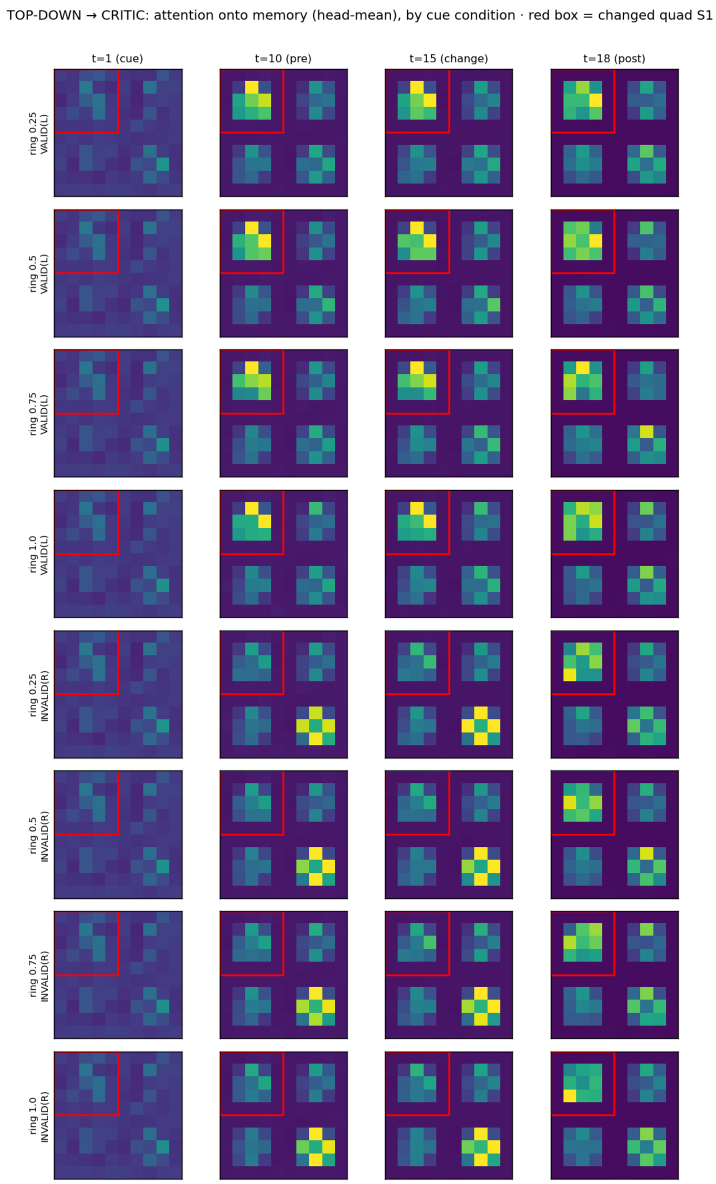

Figure 7: Top-down-stream spatial attention maps for every cue condition. The top-down read tracks the cued quadrant (S1 on valid/cue-left rows, S4 on invalid/cue-right rows) through the pre-change period — a cue-aligned prior feeding the critic.

not what distinguishes the two agents; routing is. The information is equally present in both; only the split read-out converts it into behaviour.

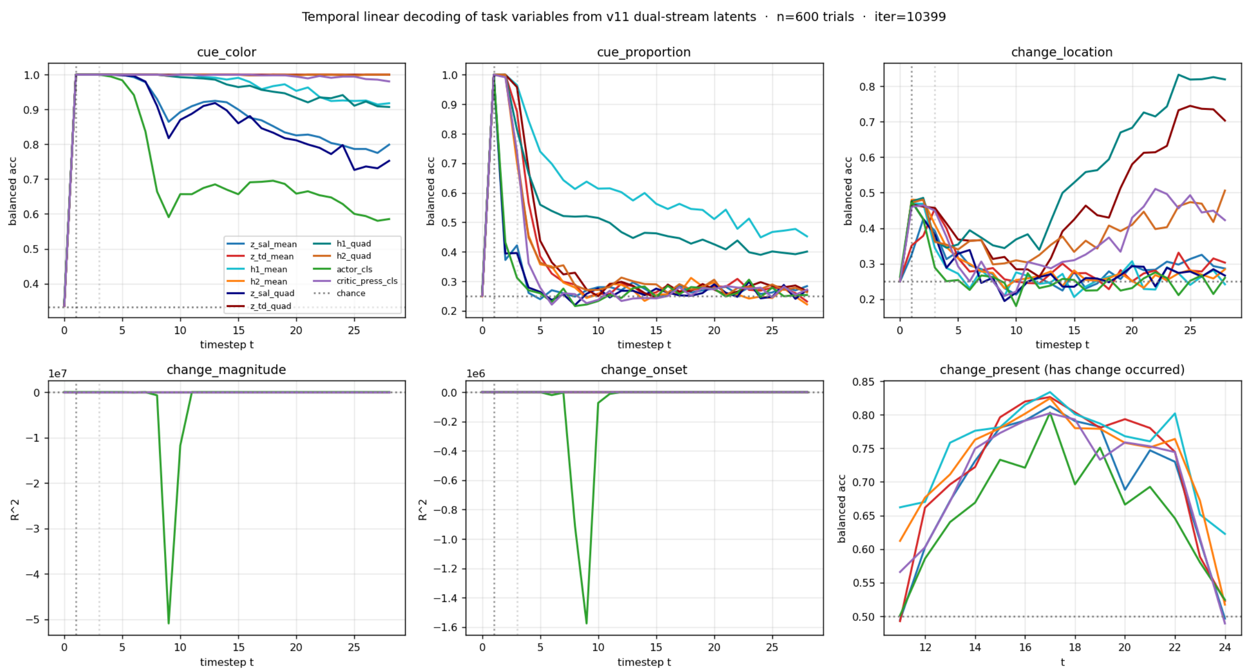

Figure 8: Temporal linear decoding of task variables from the agent's internal representations (balanced accuracy for categories, R^2 for continua; dotted lines mark chance/shuffle). Cue colour and reliability decode at ceiling from cue onset; time decodes near-perfectly; the changed location comes online after the change and is held in the per-quadrant memory.

3.4 Which stream causally carries the decision

The split read-out makes a sharp prediction: because the actor reads only the salience output and the critic reads only the top-down output, biasing salience attention should move the decision and biasing top-down attention should move value and uncertainty. We test it by injecting an additive bias into one stream's pre-softmax attention logits — artificially inflating or suppressing how strongly the image reads a chosen memory block — and re-running the greedy policy near threshold ($|\Delta\theta| = 14^\circ$, 300 trials per bias level). The prediction holds, and asymmetrically (Figures 9–10).

Top-down attention never moves the decision. Biasing the top-down keys at the changed quadrant shifts the hit rate by only Δ hit $= 0.017$ when that quadrant is cued and 0.007 when it is uncued — within noise — while moving the critic's distributional entropy substantially (Δ qent $= 0.14$ and 0.13). A uniform top-down gain is likewise inert on the decision. The stream that feeds the critic controls the value estimate and its uncertainty, not the press.

Salience attention moves the decision. Biasing the salience stream's read of the sensory memory H_1 at the changed quadrant produces the largest decision effect in the battery (Δ hit $= 0.093$) — inflating how strongly the actor's grounded stream reads the changed location makes the agent more likely to press. (Biasing the salience read of the deep memory H_2 moves the decision less, 0.043: it is the sensory trace of the change, read by salience, that the actor acts on.) Salience perturbations also move the critic's entropy — but only indirectly, because the deep memory H_2 is written from the salience output and so carries salience changes forward to the top-down stream on later frames; the reverse path does not exist, which is why top-down perturbations leave the decision untouched. The dissociation is therefore not merely quantitative but directional: salience $\longrightarrow$ decision (and, downstream, value), top-down $\longrightarrow$ value only. The split read-out installs exactly this one-way separation between the agent's choice and its valuation.

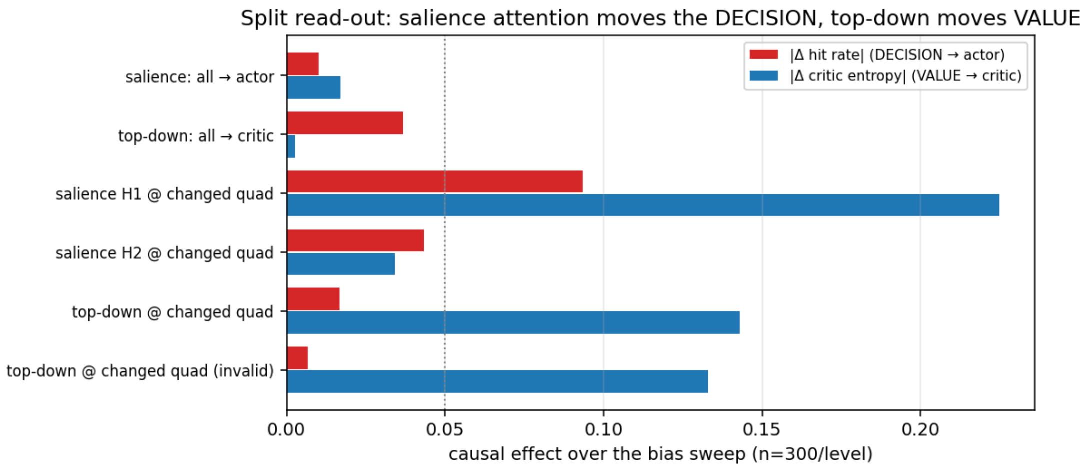

Figure 9: Causal effect of biasing each attention lever over the full sweep ($b \in [-6, 6], 300$ trials/level). Red: effect on the decision (hit rate). Blue: effect on the critic's entropy (value). Salience levers (especially the sensory-memory read at the changed quadrant) move the decision; top-down levers move value/uncertainty; uniform gains are inert.

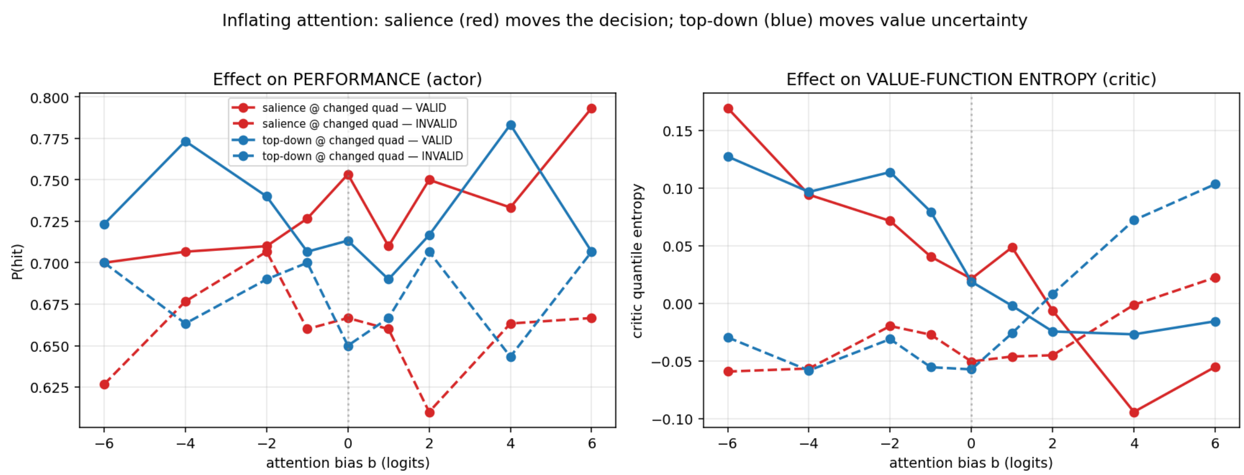

Figure 10: Inflating attention separates the actor's decision from the critic's uncertainty. Left: P(hit) versus the injected bias — the salience levers (red) move performance, the top-down levers (blue) do not. Right: the critic's quantile entropy versus the same bias — now the top-down levers (blue) dominate. Each stream controls the head it feeds.

3.5 A calibrated, value-directed critic — decoupled from the actor

The critic reads the top-down stream, and it has learned a graded value function that respects the reward magnitudes (Figure 11). Split by cue colour under a forced-wait policy, the state value rises through the trial to 3.87 for red ($v = 5$), 2.55 for green ($v = 3$) and 0.69 for blue ($v = 1$) at the change — a monotonic ordering that preserves the $5{:}3{:}1$ structure rather than collapsing the two high values together. No-change trials sit at an intermediate ≈ 2.2. The critic is reasonably calibrated against realised Monte-Carlo returns (slope 0.86, $R^2 = 0.47$ over $~$23,000 states), and its distributional spread narrows as the change magnitude grows (quantile standard deviation falling from ≈ 1.0 at $|\Delta\theta| = 0$ to ≈ 0.5 at $40°$; Figure 13).

The most informative result, however, is a disagreement between actor and critic that the split read-out makes possible. The press-versus-wait value advantage crosses zero shortly after the change for red and green (at $t = 16$ for a change at $t = 15$) but never becomes positive for blue: the critic judges that pressing on a low-value trial is not worth it. Yet the actor presses on blue anyway — recall it detects 71% of blue changes (Section 3.1). Because the actor reads the salience stream and the critic reads the top-down stream, the policy is free to detect the change regardless of its value while the critic correctly devalues it. This decoupling is exactly what lets the split-readout agent not abandon low-value trials, in contrast to a joint-readout variant whose shared representation folds the critic's value gradient into the policy and suppresses low-value detection. The split read-out separates what to do from how much it is worth.

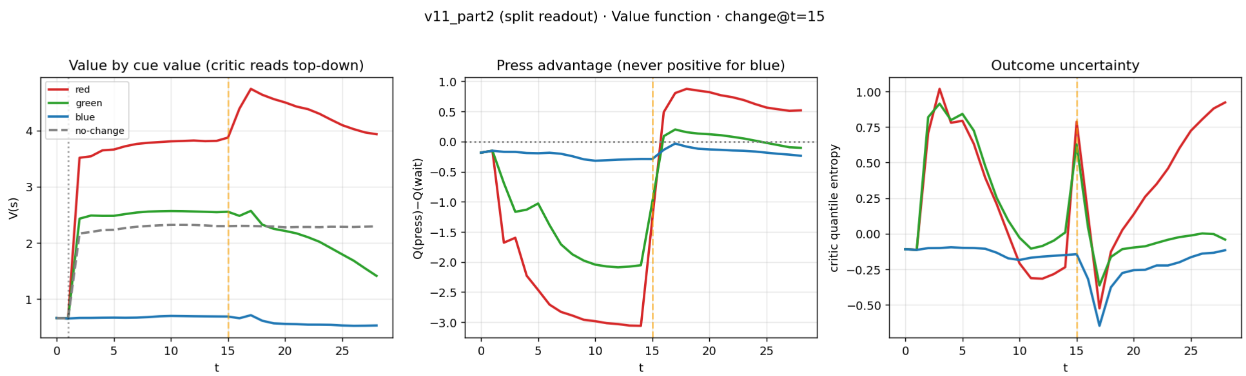

Figure 11: The distributional critic (which reads the top-down stream). Left: state value by cue colour rises through the trial in proportion to the reward at stake (red $>$ green $>$ blue). Middle: the press-versus-wait advantage crosses zero after the change for red and green but never for blue — the critic devalues pressing on low-value trials even though the actor still presses. Right: outcome uncertainty narrows with evidence.

4. Discussion

4.1 Routing, not representation, decides the behavioural phenotype

The central lesson of this agent is that where a representation is read out can matter as much as what it encodes. The agent's latents encode the cue's reliability perfectly — and so do those of a joint-readout variant that shows no cueing benefit at all. Decodability is identical; behaviour is not. The single difference is that here the grounded salience stream is wired to the actor and the top-down stream to the critic, rather than both streams feeding both heads. That routing choice switches on a reliability-scaled spatial cueing benefit, separates the agent's decision from its valuation, and installs a directional causal dissociation between the two streams. None of these are visible in what the network represents; all of them follow from how it is read.

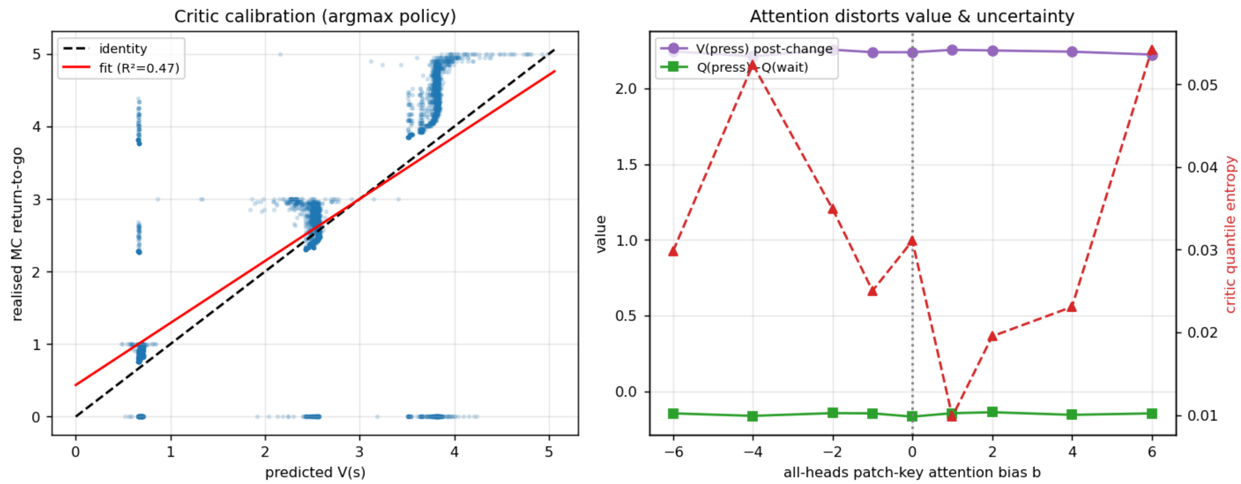

Figure 12: Critic calibration (left): predicted value versus realised Monte-Carlo return under the greedy policy. Right: the critic's distributional spread as a function of the injected top-down attention bias — a uniform top-down gain barely moves the value function, locating the critic's value computation in the spatial pattern of its attention and in the memory it reads rather than in overall top-down gain.

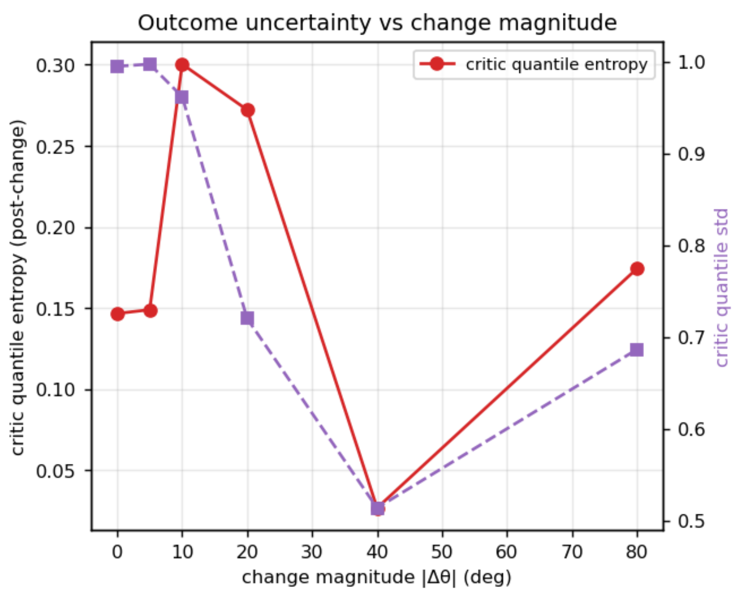

Figure 13: Outcome uncertainty versus evidence: the critic's quantile standard deviation just after the change falls as the change magnitude grows, so more detectable changes leave the value function more certain about the trial outcome.

4.2 A reliability-scaled cueing benefit from routing salience to the actor

The headline behavioural result — validly cued changes detected more accurately and faster than invalid ones, with the advantage growing from nine points at the least reliable cue to twenty-one at a fully valid one — is the canonical signature of spatial attention, and it appears here precisely because the actor reads the grounded salience stream. The salience read combines the current image with what is stored at each location across both memories; the cue's spatial prior, written into memory at cue time, is part of what the actor therefore sees, and it tips the policy toward the cued location in proportion to the prior's strength. The joint-readout variant dilutes this by averaging the salience read with the top-down read inside a shared head; the split keeps the grounded, prior-carrying read intact at the policy. The behavioural benefit is thus a direct consequence of the read-out wiring.

4.3 Decoupling choice from valuation

The split read-out also lets the agent do something the joint version cannot: detect a change well regardless of how much it is worth, while still valuing it correctly. The critic learns a graded value function that preserves the reward ordering (red $>$ green $>$ blue) and even judges that pressing on a low-value trial is not worth it — its press advantage never turns positive for blue. Yet the actor presses on blue anyway, detecting it on seven of ten trials. Because the actor and critic read different streams, the policy is not pulled down by the critic's low valuation of blue. The contrast with a joint-readout agent is instructive: when both heads share a representation, the critic's value gradient leaks into the policy and the agent abandons low-value trials; splitting the read-out frees the policy to act on perceptual evidence alone. Separating what to do from how much it is worth is, here, an architectural property rather than a learned one.

4.4 A directional causal dissociation

Perturbing the streams confirms the routing story and sharpens it into a one-way relationship. Inflating salience attention onto the changed location moves the decision; inflating top-down attention does not — it moves only the critic's value and uncertainty. The asymmetry has a mechanistic basis in the recurrence: the deep memory is written from the salience output, so salience changes propagate forward into the top-down stream (and hence into the critic), but the top-down stream writes nothing the actor reads, so its perturbations cannot reach the policy. The decision can influence value across time; value cannot influence the decision. This is the circuit-level realisation of the behavioural decoupling in Section 4.3.

4.5 Limitations and outlook

The agent is analysed at a single training snapshot and seed; it is competent but still improving, and the causal effect sizes — estimated from a few hundred trials per bias level — are modest and should be read as directional. The task is low-bandwidth (one change of one attribute in one of four locations), which both makes the clean dissociations legible and limits how hard the cue's reliability is pressed; richer tasks would test whether the reliability-scaled benefit persists when the cue must be weighed more finely. Our interventions act on attention logits, the most interpretable point; the streams also interact through the recurrent memory, and the salience-to-critic propagation we infer would be worth perturbing directly. The clean account we report — a grounded salience read routed to the actor that grants a reliability-scaled cueing benefit and decouples choice from value, and a top-down read routed to the critic that values outcomes without touching the decision — is nonetheless a concrete demonstration that read-out routing is a design lever with large, predictable behavioural consequences.

5. Conclusion

A compact recurrent vision agent solves cued, value-weighted change detection with two parallel attention streams whose outputs are split between its heads: a grounded salience stream drives the actor, a top-down gate drives the critic. This single routing choice — holding the streams, the task, and the training fixed — gives the agent a reliability-scaled spatial cueing benefit that a joint read-out does not produce, lets it detect low-value changes that a joint read-out abandons, and installs a directional causal dissociation in which salience attention moves the decision and top-down attention moves only value. The same representations, read differently, yield a different agent. Where a network is read out is itself a mechanism of attention.